

esade

Development Tools for Web and Mobile Apps

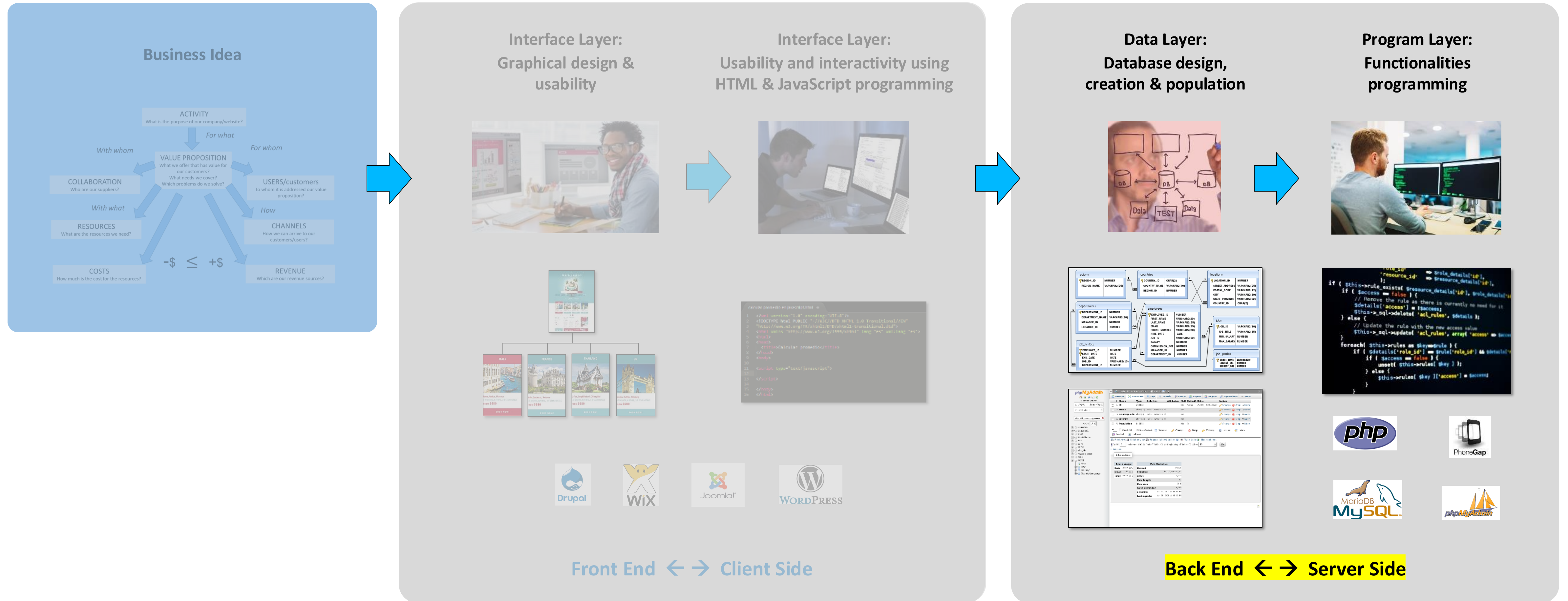
Session 5

Back End – Relational Databases & SQL

Do Good. Do Better.

esade

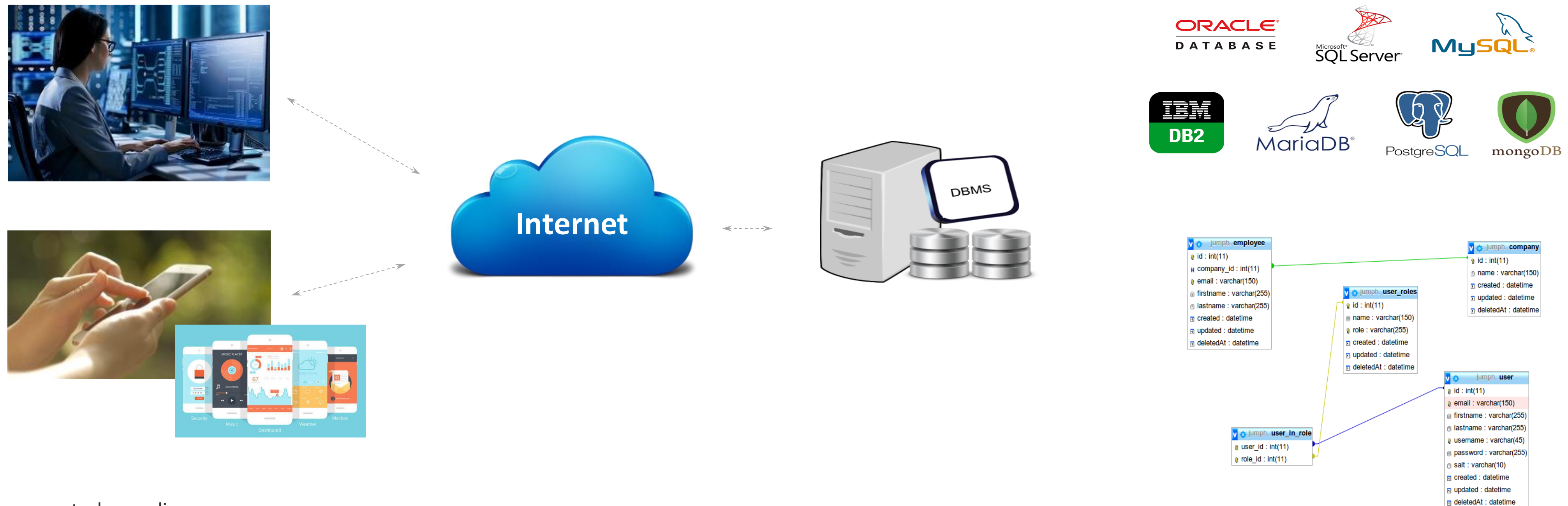
Front End Web development



What is a Database?

A **d**atab**a**se (DB) , in the most general sense, is an organized collection of data. More specifically from IT perspective, a database is an electronic system that allows data to be easily accessed, manipulated and updated (*).

DBMS: A **D**ata**B**ase **M**anagement **S**ystem is a software package designed to define, manipulate, retrieve and manage data in a database (*).



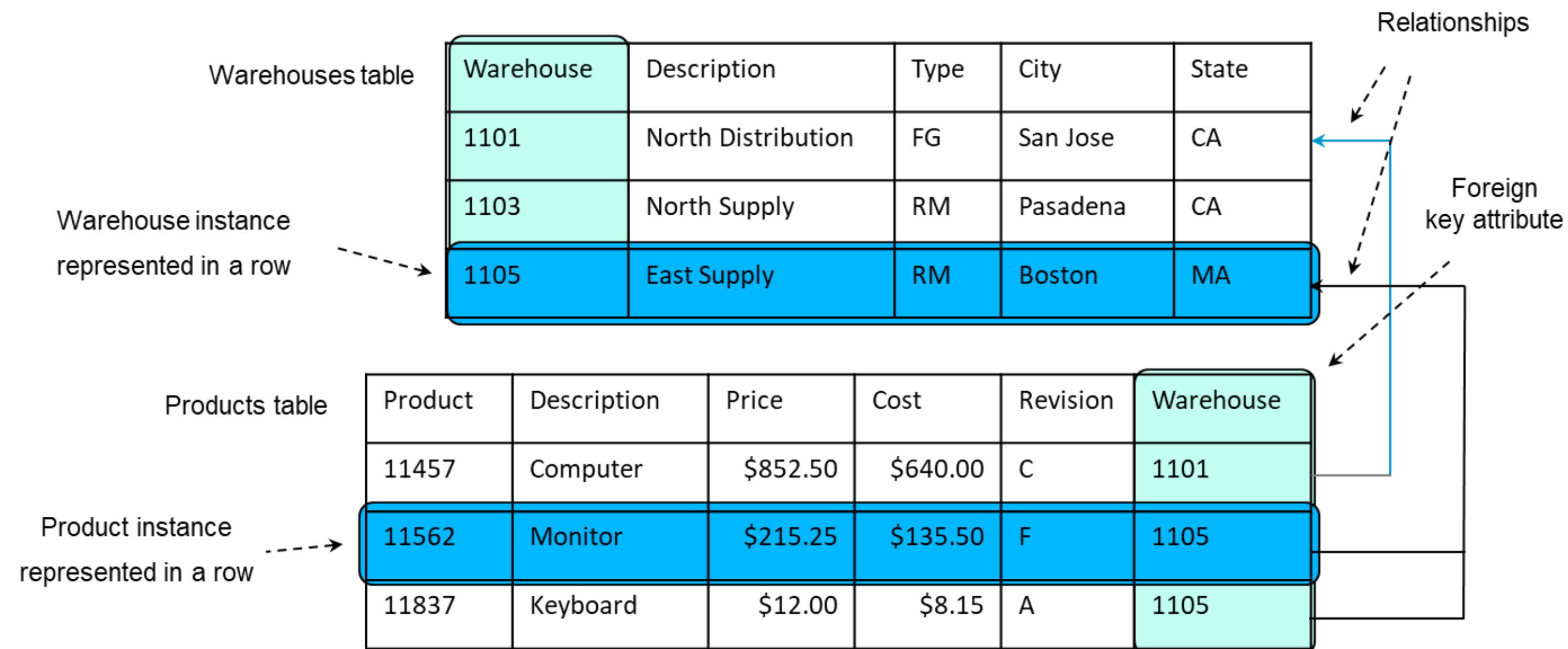
(*) Source: www.technopedia.com

Relational Databases

The relational databases, are composed of tables (relations) containing the information for each instance of the entity represented in terms of tuples (rows).

For each instance is recorded the same attribute information, corresponding to each column of the table with the same attribute for all instances (rows).

Each instance is identified by a key attribute. Tables can be related between them storing the key attribute of one table as foreign key attribute in other(s).





Deliver to Spain

All

SQL databases



Hello, Sign in
Account & Lists

Returns
& Orders



All Today's Deals Customer Service Gift Cards Registry Sell

Amazon's response to COVID-19

Books Advanced Search New Releases Best Sellers & More Children's Books Textbooks Textbook Rentals Best Books of the Month

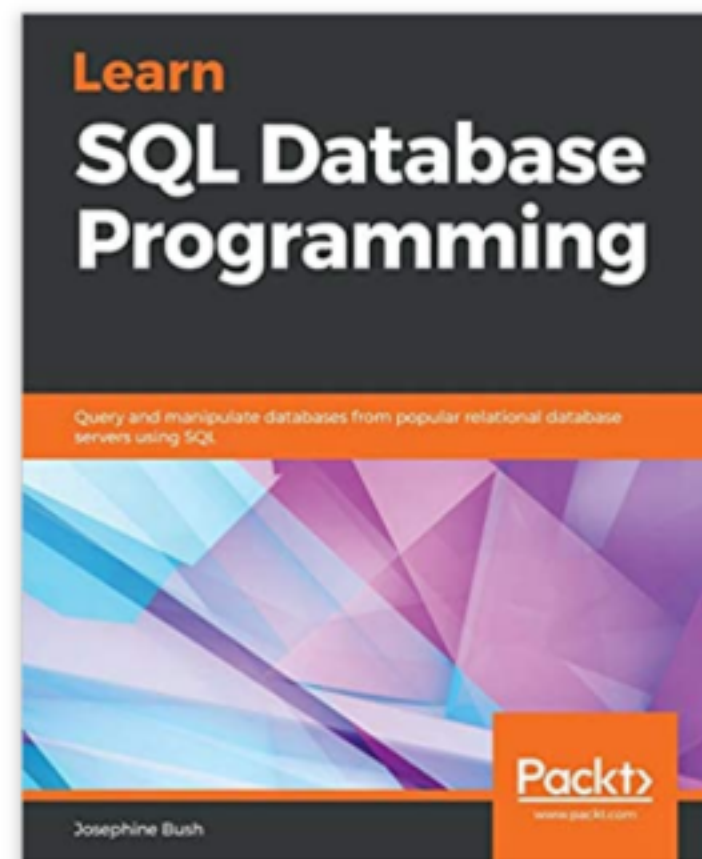
[Back to results](#)

Learn SQL Database Programming: Query and manipulate databases from popular relational database servers using SQL

by [Josephine Bush](#) (Author)

★★★★★ 8 ratings

[Look inside](#)



ISBN-13: 978-1838984762

ISBN-10: 1838984763

[Why is ISBN important?](#)

Have one to sell?

[Sell on Amazon](#)

[Add to List](#)

Share [Email](#) [Facebook](#) [Twitter](#) [Pinterest](#) [Embed](#)

amazon book clubs
early access

[Add to book club](#)

Not in a club? [Learn more](#)

Kindle \$27.99

Paperback
\$34.99 - \$39.99

Other Sellers
See all 2 versions

☐ Buy used:

\$34.99

☒ Buy new:

In Stock.

Ships from and sold by Amazon.com.

Available at a lower price from [other sellers](#) that may not offer free Prime shipping.

\$39.99

+ \$10.98 shipping

Arrives: **Thursday, July 8**

[Deliver to Spain](#)

Qty: 1

[Add to Cart](#)

[Buy Now](#)

More Buying Choices

5 New from \$39.99 | 5 Used from \$49.84

New & Used (10) from \$39.99

[See All Buying Options](#)

Available at a lower price from [other sellers](#) that may not offer free Prime shipping.

Learn everything you need to know to build efficient SQL queries using this easy-to-follow beginner's guide

Key Features

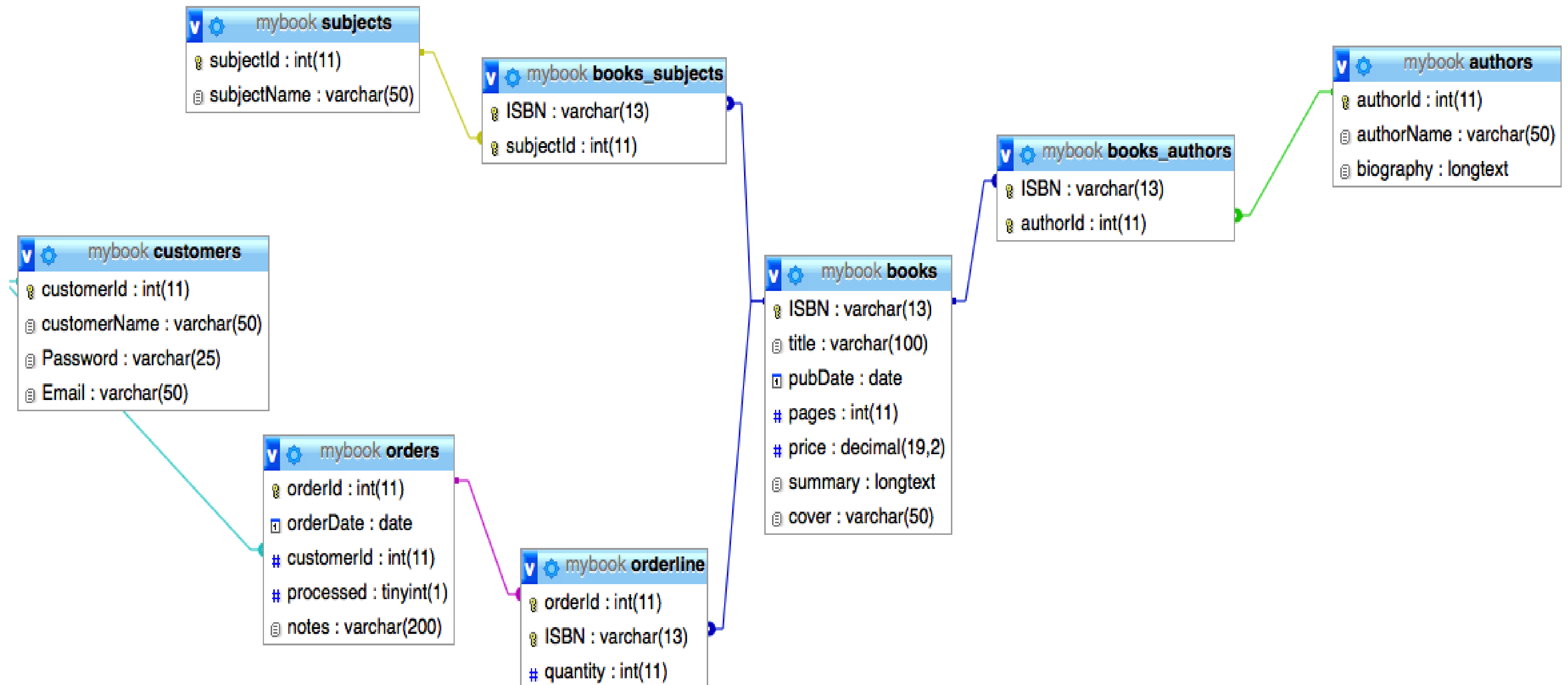
- Explore all SQL statements in depth using a variety of examples
- Get to grips with database querying, data aggregate, manipulation, and much more

[Read more](#)

Database Schema

MyBook is a simulated e-commerce bookstore designed and used at ESADE for teaching purposes.

Schema of MyBook database:





TRAVEL ▾

FLYING INFO ▾

Have any questions?



 **vueling** CLUB

Log in

Sign up

 Vueling update: [Flexibility for your bookings](#) | [Travel restrictions and guidance](#) | [Hand luggage policy](#)

 FLIGHTS

 CARS

 HOTELS

 BE INSPIRED

 ONLINE CHECK-IN

 YOUR BOOKING

Origin

BCN Barcelona

Destination

LON London

Outbound

21/06/2021 

Return

28/06/2021 

1 Adult, 0 Children, 0 Infants

1 passenger ▾

Search

>

- ALREADY VACCINATED?

- ¡SÍ!

TRAVEL TO SPAIN JUST GOT EASIER

Book now

>



 Be Inspired

Find your ideal destination

Flights

esade

Booking is the table of Flights Database to be used for the SQL examples. This is a hyper simplified database composed by a single table(*).

bookingid	bookingdate ▾ 1	firstname	lastname	email	gender	country	region	postalcode	amount	airline	seatclass
KF1751	2017-12-27 00:00:00	Issiah	Cocker	icockerh@jigsy.com	Male	France	Bretagne	22091CEDEX9	1774.65	Lufthansa	First Class
PC5186	2017-12-23 00:00:00	Cristian	Brockherst	cbrockherst5q@examiner.com	Male	France	Poitou-Charentes	86206CEDEX	2917.64	Iberia	Grand Class
CR7468	2017-12-22 00:00:00	Robinia	Hryniewicz	rhryniewicz3l@harvard.edu	Female	France	Île-de-France	94593CEDEX2	207.55	Delta Airlines	Economy
WK8140	2017-12-22 00:00:00	Angie	Matelaitis	amatelaitis7s@sohu.com	Male	France	Lorraine	54412CEDEX	1939.39	Delta Airlines	First Class
UW2664	2017-12-20 00:00:00	Biddie	Reany	breany1j@home.pl	Female	France	Basse-Normandie	50109CEDEX	1005.87	Air France	First Class
SE1354	2017-12-20 00:00:00	Maryellen	Petry	mpetry4s@baidu.com	Female	France	Île-de-France	78942CEDEX	1432.29	Vueling	First Class
FP3399	2017-12-19 00:00:00	Jania	Sigars	jsigars3g@usa.gov	Female	Germany	Baden-Württemberg	79106000000	686.29	Air France	Business
KP9286	2017-12-14 00:00:00	Jamesy	Champken	jchampken18@washingtonpost.com	Male	France	Auvergne	15015CEDEX	1364.36	Air France	First Class
TQ3079	2017-12-14 00:00:00	Ofella	Feldklein	ofeldklein6p@deviantart.com	Female	Germany	Berlin	13509000000	273.61	Air France	Economy
AB0730	2017-12-11 00:00:00	Clementius	Pyrton	cpyrton3z@dot.gov	Male	France	Île-de-France	92504CEDEX	2720.03	Iberia	Grand Class
EW3550	2017-12-09 00:00:00	Cosmo	Nollet	cnollet5u@utexas.edu	Male	France	Rhône-Alpes	69424CEDEX0	2447.86	RyanAir	Grand Class
XP5286	2017-12-06 00:00:00	Mortie	Salomon	msalomon4c@nifty.com	Male	France	Haute-Normandie	76418CEDEX9	782.41	Air France	Business
WR6518	2017-12-05 00:00:00	Cthrine	MacIlraith	cmacilraith5v@geocities.jp	Female	France	Île-de-France	78804CEDEX	307.76	Air France	Economy
PC2138	2017-12-04 00:00:00	Margy	Venditto	mvenditto4v@utexas.edu	Female	France	Picardie	80075CEDEX1	307.53	Vueling	Economy
GQ2328	2017-12-02 00:00:00	Farrand	Dhennin	fdhennin46@oaic.gov.au	Female	Netherlands	Provincie Zuid-Holland	3009000000	423.46	Iberia	Economy
MU9033	2017-12-01 00:00:00	Ogdan	Wapol	owapol32@nature.com	Male	France	Franche-Comté	25014CEDEX	1601.42	American Airlines	First Class
KQ8915	2017-11-19 00:00:00	Thayne	Claiton	tcclaiton7x@gmpg.org	Male	France	Lorraine	54304CEDEX	34.67	RyanAir	Economy
SB6644	2017-11-06 00:00:00	Hank	Drache	hdrache6f@cmu.edu	Male	Netherlands	Provincie Gelderland	7004000000	325.17	EasyJet	Economy
EO5197	2017-11-02 00:00:00	Selinda	Spedroni	sspedroni2f@home.pl	Female	Netherlands	Provincie Zuid-Holland	3009000000	1797.04	Air France	First Class
GM4725	2017-11-02 00:00:00	Chastity	Stilwell	cstilwell2o@parallels.com	Female	Netherlands	Provincie Overijssel	8029000000	963.94	EasyJet	Business
NS2904	2017-10-25 00:00:00	Felix	Karpenya	fkarpenya7l@blog.com	Male	Germany	Sachsen	1462000000	131.23	Iberia	First Class
AE1479	2017-10-22 00:00:00	Faber	Calderon	fcalderson1d@typepad.com	Male	France	Centre	45933CEDEX9	2510.68	RyanAir	Grand Class
OP8009	2017-10-21 00:00:00	Myrah	Machans	mmachans7w@yahoo.co.jp	Female	France	Champagne-Ardenne	51715CEDEX	221.25	Lufthansa	Grand Class
NR4102	2017-10-19 00:00:00	Shanie	Daughtrey	sdaughtrey68@goo.gl	Female	France	Île-de-France	78424CEDEX	2147.35	Lufthansa	Grand Class
DZ3839	2017-10-17 00:00:00	Anatole	Luckhurst	aluckhurst2d@stumbleupon.com	Male	France	Île-de-France	75434CEDEX0	1327.48	Air France	First Class

(*) This is not a realistic database. Its purpose is purely academic, to present SQL examples working with a single table.

SQL: Insert, Update, Delete

Insert instruction:

```
INSERT INTO Table (Attribute 1, Attribute 2, ..., Attribute N)  
VALUES (Value 1, Value 2, ..., Value N)
```

Update instruction:

```
UPDATE Table  
SET Attribute = Value  
[WHERE Condition] ← Optional clause
```

Delete instruction:

```
DELETE FROM Table  
[WHERE Condition] ← Optional clause
```

SQL: Insert, Update, Delete examples

- Insert into Flights table a new row.

```
INSERT INTO booking (bookingid, bookingdate, firstname, lastname, email, gender, country, region,  
                    postalcode, amount, airline, seatclass)  
VALUES ("GW0247", "2020-08-14", "Bjorn", "Olson", "willol@gmail.com", "Male", "Norway", , , 440.5,  
        "Delta Airlines", "Business")
```

- Due to an error introducing the data in Flights table, the Economy class seats are 10% below their real price. Update Flights table increasing 10% the price of all the records for this type of seat (Economy).

```
UPDATE booking  
SET amount = amount * 1.1  
WHERE seatclass = "Economy";
```

- Delete the record with bookingid UW2664. It was incorrectly created.

```
DELETE FROM booking  
WHERE bookingid = "UW2664"
```

SQL: Select

Select instruction (simplified version):

SELECT * | List of Attributes

FROM Table

[**WHERE** Condition] ← Optional clause

[**ORDER BY** Attribute] ← Optional clause

SQL: Select examples

- List all the clients that flew with Delta Airlines.

```
SELECT firstname, lastname  
FROM booking  
WHERE airline = "Delta Airlines";
```

firstname	lastname
Dion	Dikle
Wes	Wattam
Orsa	Patterson
Broddie	Oppery
Sandi	Furzer
Adah	Tonepohl
Deena	Melladew
Gaspard	Furniss
Marijn	Gredden
Jorey	Uren
Trumann	Merriott
Caprice	Skey
Garrick	Finlayson
Daniela	Lipscombe
Caro	Ferrarese
Robinia	Hryniewicz
Balduin	Allabush
Godwin	Mileham
Arlen	Beresford
Morton	Coveley
Cece	Dominelli
Leroi	De Carolis
Bee	Breitler
Ibbie	Partner
Kare	Follitt

SQL: Select examples

- List all the German clients that flew with RyanAir.

SELECT firstname, lastname, country, airline

FROM booking

WHERE country = “Germany” AND airline = “RyanAir”;

firstname	lastname	country	airline
Remington	Thornewill	Germany	RyanAir
Nicolette	Chamberlaine	Germany	RyanAir
Yulma	Clue	Germany	RyanAir
Brigham	Kubis	Germany	RyanAir

SQL: Select examples

- List all the clients from Spain and United Kingdom.

```
SELECT firstname, lastname, country  
FROM booking  
WHERE country = "Spain" OR country = "United Kingdom";
```

firstname	lastname	country
Pascale	Finder	United Kingdom
Karlyn	Laverick	United Kingdom
Flora	Duckit	Spain
Orsa	Patterson	Spain
Rae	Mowsdell	Spain
Marijn	Gredden	Spain
Travus	Sackey	Spain
Arte	Lehrmann	Spain
Benedict	Asee	United Kingdom
Olvan	Baudet	Spain
Kaleb	Norley	United Kingdom
Dot	Hutcheon	United Kingdom
Winny	Maro	United Kingdom
Xymenes	Sunnucks	United Kingdom
Audrye	Christopher	United Kingdom
Xenos	Dunbabin	Spain
Fergus	Snoddy	Spain
Barrie	Wakerley	Spain

SQL: Select examples

- List all the clients from France that flew on Air France or RyanAir.

```
SELECT firstname, lastname, country, airline
FROM booking
WHERE country = "France" AND
      (airline = "Air France" OR airline = "RyanAir");
```

firstname	lastname	country	airline
Catlee	Sponder	France	Air France
Ulises	Firth	France	RyanAir
Giacopo	Haffenden	France	Air France
Parsifal	Loveguard	France	RyanAir
Jamesy	Champken	France	Air France
Millisent	Habert	France	RyanAir
Faber	Calderon	France	RyanAir
Michael	Querree	France	Air France
Lazare	Plumbley	France	Air France
Imelda	Grzeskowski	France	RyanAir
Ervin	Noore	France	Air France
Biddie	Reany	France	Air France
Klement	Ponting	France	RyanAir
Ivonne	Murr	France	Air France
Tiffi	Allitt	France	RyanAir
Lezlie	Pymm	France	RyanAir
Mickey	Van Zon	France	RyanAir
Petra	Miroy	France	RyanAir
Fabio	Fendlow	France	RyanAir
Phil	Stubs	France	RyanAir
Hamil	Hogsden	France	Air France
Anatole	Luckhurst	France	Air France
Chrissie	Walshaw	France	Air France
Israel	Orwin	France	Air France
Ernaline	Edon	France	Air France

SQL: Select with aggregated functions

Select instruction with aggregated functions:

```
SELECT Count(Att) | Sum(Att) | Avg(Att) | Min(Att) | Max(Att)
FROM Table
[WHERE Condition]      ← Optional clause
```

Calculation performed for each aggregated function:

- Count(*) → **Counts** all rows given by the Select instruction
- Sum(NumericAtt) → Calculates the **Summa** of NumericAtt for all rows given by the Select
- Avg(NumericAtt) → Calculates the **Average** of NumericAtt for all rows given by the Select
- Min(NumericAtt) → Calculates the **Smallest** value of NumericAtt in all rows given by the Select
- Max(NumericAtt) → Calculates the **Largest** value of NumericAtt in all rows given by the Select

SQL: Select with aggregated functions examples

- How many flight tickets were sold in 2017 ?.

```
SELECT count(*) AS Number_of_tickets_sold  
FROM booking  
WHERE bookingdate >= "2017-01-01" AND bookingdate <= "2017-12-31";
```

```
SELECT count(*) AS Number_of_tickets_sold  
FROM booking  
WHERE year(bookingdate) = 2017;
```

Number_of_tickets_sold

SQL: Select with aggregated functions examples

- How much is the amount of all flight tickets sold in August 2017 ?.

```
SELECT sum(amount) AS Amount_Aug_2017  
FROM booking  
WHERE year(bookingdate) = 2017 AND month(bookingdate) = 8;
```

Amount_Aug_2017

15116.63

SQL: Select with aggregated functions examples

- How much costed the cheapest and the most expensive flight tickets sold in 2017 ?.

```
SELECT min(amount) AS "Cheapest ticket amount 2017", max(amount) AS  
"Most expensive ticket amount 2017",  
FROM booking  
WHERE year(bookingdate) = 2017;
```

Cheapest ticket amount 2017

17.90

Most expensive ticket amount 2017

2963.38

esade

SQL: GROUP BY



idreserva	fechareserva	nombre	apellido	email	genero	Pais	region	codpostal	Importe	compania	clase	pais_origen	pais_destino
NY0506	2015-11-24 00:00:00	Harper	O'Cuddie	hocuddie0@yandex.ru	Male	Germany	Baden-Württemberg	76199000000	1914.49	EasyJet	First Class	Canada	UAE
AM8751	2017-02-05 00:00:00	Sigrid	Zelner	szelner1@liveinternet.ru	Female	France	Pays de la Loire	44019CEDEX1	1045.07	American Airlines	First Class	Mexico	USA
A07709	2016-05-11 00:00:00	Harcourt	Haddy	hhaddy2@addthis.com	Male	France	Île-de-France	91309CEDEX	103.92	Iberia	First Class	Germany	France
FS7074	2015-09-08 00:00:00	Dion	Dikle	ddikle3@businessinsider.com	Female	France	Île-de-France	75954CEDEX1	2558.01	Delta Airlines	Grand Class	Russia	Germany
RC1709	2017-10-09 00:00:00	Pascale	Finder	pfinder4@ebay.co.uk	Male	United Kingdom	England	DL10	560.89	American Airlines	Business	Thailand	Canada
AX7724	2015-06-18 00:00:00	Mandel	Atlee	matlee5@cargocollective.com	Male	France	Languedoc-Roussillon	34080000000	473.94	Lufthansa	Economy	United Kingdom	Norway
JH4675	2017-03-09 00:00:00	Catlee	Sponder	csponder6@accuweather.com	Female	France	Île-de-France	77504CEDEX	2167.93	Air France	Grand Class	Spain	France
QC6505	2017-08-18 00:00:00	Ulises	Firth	ufirth7@nyu.edu	Male	France	Île-de-France	94309CEDEX	504.65	RyanAir	Business	Russia	Italy
HS0785	2016-08-19 00:00:00	Drucill	Bremmell	dbremmell8@independent.co.uk	Female	France	Île-de-France	75756CEDEX1	1888.25	Vueling	First Class	Mexico	Japan
KC6386	2017-06-25 00:00:00	Alec	Gasgarth	agasgarth9@blogger.com	Male	France	Île-de-France	92024CEDEX	1870.14	American Airlines	First Class	USA	Argentina
PU9173	2015-09-13 00:00:00	Karlyn	Laverick	klavericka@pbs.org	Female	United Kingdom	England	NR34	2602.23	EasyJet	Grand Class	USA	Norway
QP4523	2016-01-24 00:00:00	Melisandra	Toothill	mtoothillb@wix.com	Female	France	Haute-Normandie	76804CEDEX	2725.23	American Airlines	Grand Class	Spain	Netherlands
HR7714	2016-05-30 00:00:00	Wes	Wattam	wwattamc@godaddy.com	Male	France	Nord-Pas-de-Calais	59425CEDEX	2244.95	Delta Airlines	Grand Class	Chile	USA
SJ3807	2017-01-15 00:00:00	Flora	Duckit	fduckitd@umich.edu	Female	Spain	Canarias	35213000000	1149.33	EasyJet	First Class	USA	Singapur
C04027	2017-05-25 00:00:00	Phillie	Schwandt	pschwandte@tuttocitta.it	Female	France	Bretagne	56019CEDEX	1218.79	Iberia	First Class	Spain	China
AF9669	2017-05-19 00:00:00	Orsa	Patterson	opattersonf@walmart.com	Female	Spain	Galicia	36205000000	1052.22	Delta Airlines	First Class	USA	UAE
FB7396	2016-03-09 00:00:00	Jennilee	Sherwill	jsherwillg@about.me	Female	France	Île-de-France	94525CEDEX1	274.86	Iberia	Economy	United Kingdom	Norway
KF1751	2017-12-27 00:00:00	Issiah	Cocker	icockerh@jigsy.com	Male	France	Bretagne	22091CEDEX9	1774.65	Lufthansa	First Class	Mexico	USA
YD1344	2016-03-24 00:00:00	Broddie	Oppery	bopperyi@biblegateway.com	Male	France	Pays de la Loire	53104CEDEX	2943.25	Delta Airlines	Grand Class	Singapur	Norway
SY9287	2015-04-03 00:00:00	Sandi	Furzer	sfurzerj@fc2.com	Female	Netherlands	Provincie Friesland	8914000000	453.61	Delta Airlines	Economy	Norway	Netherlands
KS0894	2016-06-25 00:00:00	Orv	Whitehead	owhiteheadk@pbs.org	Male	France	Centre	37554CEDEX	558.08	Lufthansa	Business	Russia	China
OX7708	2017-09-25 00:00:00	Olav	MacNelly	omacnellyl@sciencedirect.com	Male	Italy	Piemonte	10141000000	239.28	Lufthansa	Grand Class	Argentina	Peru
NR9943	2017-04-01 00:00:00	Levi	Kincade	lkincadem@samsung.com	Male	Germany	Hessen	60325000000	1871.66	Air France	First Class	United Kingdom	Norway
HF0178	2015-09-06 00:00:00	Bess	Southam	bsouthamn@imageshack.us	Female	Netherlands	Provincie Flevoland	1354000000	2939.66	Air France	Grand Class	United Kingdom	Russia
Z02208	2017-07-06 00:00:00	Gilbertine	Lant	glanto@digg.com	Female	France	Franche-Comté	25086CEDEX9	881.52	Vueling	Business	Canada	Thailand
DR8319	2016-11-08 00:00:00	Philipa	Alabone	palabonep@shareasale.com	Female	Netherlands	Provincie Noord-Hol...	1094000000	1629.26	Lufthansa	First Class	Japan	Peru
HP0626	2016-03-19 00:00:00	Abie	Albarez	aalbarezq@npr.org	Male	France	Rhône-Alpes	42304CEDEX	2368.67	EasyJet	Grand Class	Mexico	USA
NA9206	2017-04-03 00:00:00	Kerwinn	Zamora	kzamorar@cafepress.com	Male	Netherlands	Provincie Flevoland	8214000000	385.49	Vueling	Economy	United Kingdom	France
FG9657	2016-11-16 00:00:00	Camilla	Cranke	ccrankes@wisc.edu	Female	France	Île-de-France	93335CEDEX	2202.82	EasyJet	Grand Class	Mexico	USA
PQ2203	2015-09-02 00:00:00	Timothy	Cornuau	tcornuaut@omniture.com	Male	France	Poitou-Charentes	16919CEDEX9	1210.44	American Airlines	First Class	Canada	France
IJ7028	2015-08-25 00:00:00	Parnnell	Briar	pbriaru@princeton.edu	Male	Netherlands	Provincie Noord-Hol...	1814000000	2490.14	American Airlines	Grand Class	Russia	Japan
WL5455	2017-03-20 00:00:00	Consalve	Gilluley	cgilluleyv@domainmarket.com	Male	Italy	Lazio	185000000	881.17	Air France	Business	United Kingdom	France

SQL: Complete SELECT instruction

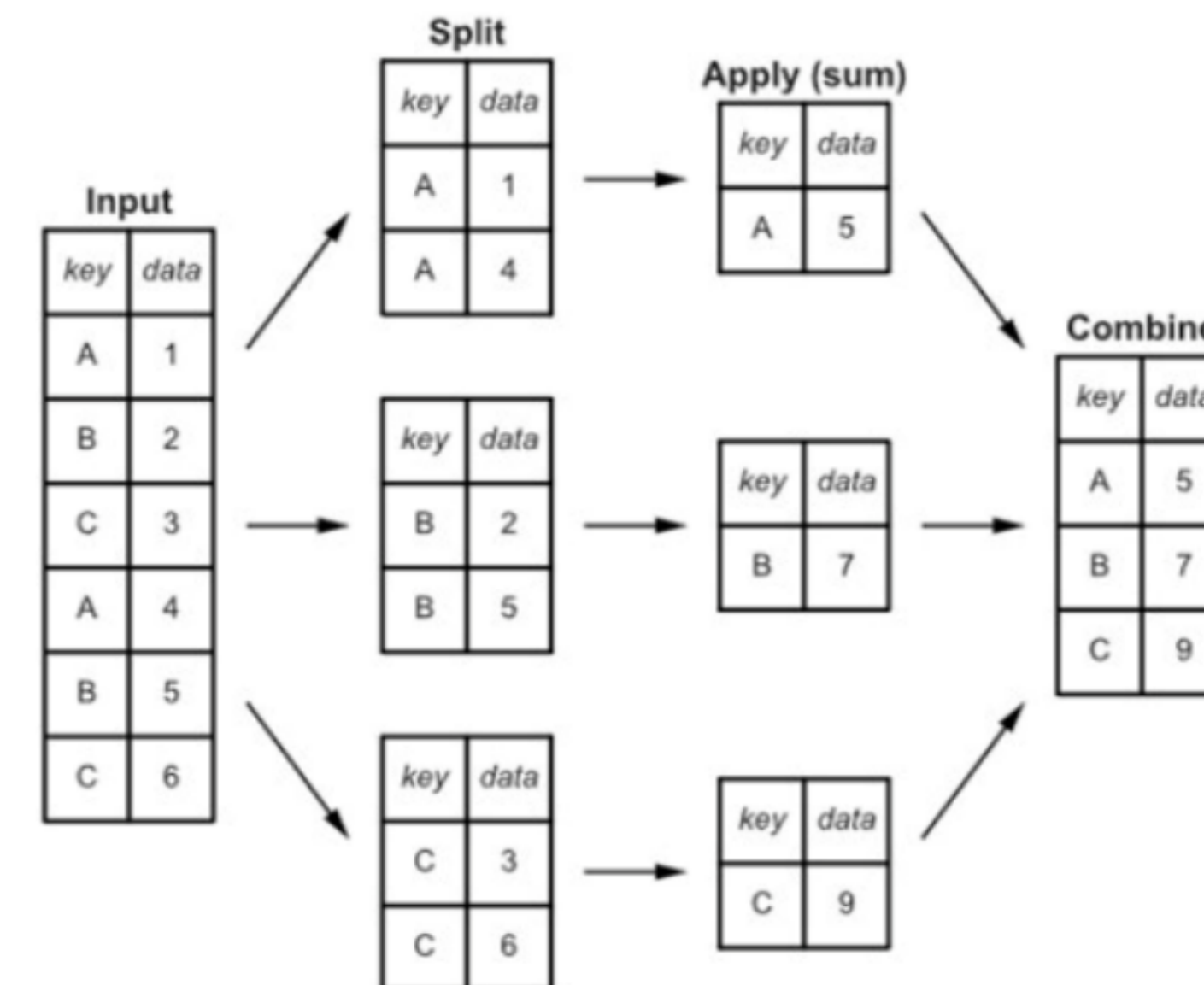
```
SELECT [DISTINCT] attribute_list | *  
FROM table  
[WHERE constraint_expression ]  
[GROUP BY attribute_list]  
[HAVING constraint_expression_on_groups]  
[ORDER BY column ASC/DESC ]  
[LIMIT count];
```

SQL: Grouping rows

- The purpose is the grouping by attribute values, not by the entire table.
- Generally, the grouping has associated the computation of the value of an aggregate function, although possibly only just want to know the existence of the groups.

- The GROUP BY uses the split-apply-merge strategy for data analysis

- In the division phase, it divides the groups with their values.
- In the apply phase, it applies the aggregation function and outputs a single value.
- In the merge phase, you merge the groups with values into a single value.



SQL: Grouping rows

codpostal	Importe	compania	clase
76199000000	1914.49	EasyJet	First Class
44019CEDEX1	1045.07	American Airlines	First Class
91309CEDEX	103.92	Iberia	First Class
75954CEDEX1	2558.01	Delta Airlines	Grand Class
DL10	560.89	American Airlines	Business
34080000000	473.94	Lufthansa	Economy
77504CEDEX	2167.93	Air France	Grand Class
94309CEDEX	504.65	RyanAir	Business
75756CEDEX1	1888.25	Vueling	First Class
92024CEDEX	1870.14	American Airlines	First Class
NR34	2602.23	EasyJet	Grand Class
76804CEDEX	2725.23	American Airlines	Grand Class
59425CEDEX	2244.95	Delta Airlines	Grand Class
35213000000	1149.33	EasyJet	First Class
56019CEDEX	1218.79	Iberia	First Class
36205000000	1052.22	Delta Airlines	First Class
94525CEDEX1	274.86	Iberia	Economy
22091CEDEX9	1774.65	Lufthansa	First Class
53104CEDEX	2943.25	Delta Airlines	Grand Class
8914000000	453.61	Delta Airlines	Economy
37554CEDEX	558.08	Lufthansa	Business
10141000000	239.28	Lufthansa	Grand Class
60325000000	1871.66	Air France	First Class
1354000000	2939.66	Air France	Grand Class
25086CEDEX9	881.52	Vueling	Business
1094000000	1629.26	Lufthansa	First Class

Number of bookings per airline, sorted out by number of bookings

```
SELECT booking.airline, count(*) AS bookings_per_airline
FROM booking
GROUP BY booking.airline
ORDER BY count(*);
```

The table rows are grouped by airline, then counting how many rows there are in each group and finally sorting the groups ascending

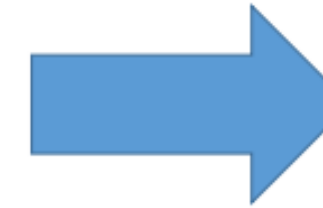


airline	bookings_per_airline
RyanAir	115
Delta Airlines	115
American Airlines	127
Lufthansa	127
EasyJet	127
Air France	129
Iberia	129
Vueling	131

SQL: Grouping rows

How many bookings there are of each seat class?
Sort out ascending by number of bookings.

```
SELECT booking.seatclass, count(*) AS bookings_per_seatclass
FROM booking
GROUP BY booking.seatclass
ORDER BY count(*);
```



seatclass	bookings_per_seatclass
Economy	110
Business	180
First Class	354
Grand Class	356

How many bookings there are of each seat class?
Show only those seat classes with more than 200 bookings.
Sort out ascending by number of bookings.

```
SELECT booking.seatclass, count(*) AS bookings_per_seatclass
FROM booking
GROUP BY booking.seatclass
HAVING count(*) > 200
ORDER BY count(*);
```



seatclass	bookings_per_seatclass
First Class	354
Grand Class	356

A dark blue diagonal banner runs across the top of the image, sloping downwards from left to right.

esade

Do Good. Do Better.